

Hyperparameter Tuning & Model Evaluation Report

Machine Learning for Gunshot Residue Classification:

Accuracy, Interpretability, and Failure Analysis

Hyperparameter Tuning & Model Evaluation Report

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04/25/2026

Prepared for:

National Institute of Justice (U.S. Department of Justice)

1. Introduction

When a firearm is discharged, microscopic particles containing characteristic combinations of the three key elements lead (Pb), barium (Ba), and antimony (Sb) are deposited on nearby surfaces and persons. These gunshot residue (GSR) particles are collected and analyzed using scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM/EDS) as physical evidence in criminal proceedings. But deciding whether a given particle is truly GSR or an environmental lookalike remains a bottleneck heavily dependent on manual analysis. It depends on expert judgment, is slow to scale, and is dangerously susceptible to error in both directions.

A false positive, for example labeling brake dust, firework residue, or industrial debris as GSR, can place an innocent person at a crime scene. A false negative, on the other hand, failing to identify genuine GSR, can let a guilty party walk. Barium-containing environmental particles can routinely mimic GSR signatures, and single-element particles occupy an inherently ambiguous space where even experienced analysts disagree. This asymmetric but universally high cost of misclassification in either direction makes GSR classification a domain where model accuracy alone is an insufficient metric. What matters equally is understanding where, why, and how often a model gets it wrong.

This report applies machine learning methods to the GSR classification problem not just to automate it, but to interrogate it. We train three model architectures, logistic regression, XGBoost, and a multi-layer perceptron neural network, on 2.3 million labeled particles from the Netherlands Forensic Institute (NFI), and we evaluate them

not only on aggregate performance but on failure behavior: which particle classes produce false positives, how the models handle ambiguous single-element particles they were never trained on, and whether near-perfect accuracy masks meaningful vulnerabilities at the decision boundary. The goal is not to build a black box that replaces forensic analysts, but to build a transparent, failure-aware tool that tells practitioners exactly when it should and should not be trusted.

The report is organized as follows: we cover the research question and hypothesis; data acquisition and preprocessing; exploratory data analysis; feature engineering by model type; baseline modeling results with failure analysis; final models with tuned hyperparameters; and a discussion of caveats, limitations, and next steps.

Background and Research Question

The identification of GSR particles is a cornerstone of forensic investigations involving firearm discharge. When a firearm is fired, microscopic particles containing characteristic elements such as lead (Pb), barium (Ba), and antimony (Sb) are deposited on nearby surfaces and can serve as physical evidence in criminal proceedings. Traditional GSR analysis relies on scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM/EDS), a process that depends heavily on expert judgment and is susceptible to both false positives from chemically similar environmental particles and false negatives from atypical GSR compositions.

Redefined Research Question

How accurately and reliably can machine learning models distinguish true gunshot residue particles from chemically similar non-GSR particles, and what are the limits of these models when applied to ambiguous single element particles? We also examine which features drive predictions and where models fail, particularly on ambiguous particle types.

2. Redefined Hypothesis & Prediction

We hypothesize that machine learning models can effectively distinguish true GSR particles from environmental confounders, when the classic multi-element signatures (Pb-Ba-Sb combinations) are present, achieving near-perfect classification on the binary task, with non-linear models (e.g., gradient-boosted trees, neural networks) slightly outperforming linear models (e.g., logistic regression). We further hypothesize that single element particles will lead to substantial uncertainty in model classifications with degraded performance compared to multi element particles. This reflects genuine forensic uncertainty, or aleatoric uncertainty.

We expect misclassifications to concentrate among barium-heavy non-GSR particles and ambiguous single-element particles, highlighting the limits of automated classification.

3. Methods & Results

The following sections give an overview of the fitting, hyperparameter tuning, and evaluation process for each baseline model as well as adjustments made from the baseline models outlined in the baseline models report.

Hyperparameter Tuning & Model Optimization

Logistic regression

The logistic regression model was selected as the baseline linear classifier due to its interpretability and ability to provide direct insight into feature importance through coefficient estimates. Unlike the neural network and XGBoost models, logistic regression serves as a transparent benchmark, allowing us to evaluate how well a linear decision boundary can separate GSR from Non-GSR particles.

Initial baseline results indicated near-perfect separability when all elemental features were included, driven primarily by the dominant influence of lead (Pb), barium (Ba), and antimony (Sb). To better assess realistic model performance and avoid overly deterministic classification, these three elemental features were removed prior to tuning. This forces the model to rely on broader elemental patterns and engineered features rather than the most obvious GSR indicators.

The model pipeline consisted of median imputation, standard scaling, and logistic regression with class weighting set to “balanced” to account for class imbalance. A group-aware cross-validation strategy using GroupKFold was implemented to prevent leakage between particles originating from the same SEM stub.

Hyperparameter tuning was conducted using a randomized search over the following parameters:

- **Regularization strength (“C”)**: Controls model complexity
- **L1 Ratio**: Governs the balance between L1 (features selection) and L2 (coefficient shrinkage) regularization
- **Class Weight**: Fixed at “balanced”

The best-performing configuration achieved a cross-validation ROC-AUC of approximately 0.906, with optimal parameters:

- **Regularization**: Elastic Net
- **C**: 1.76
- **L1 Ratio**: 0.21
- **Class Weight**: balanced

The relatively low L1 ratio indicates that the model favors L2 regularization, suggesting that predictive signal is distributed across multiple features rather than concentrated in a sparse subset.

XGB

To finalize the XGBoost model, several steps were taken to solidify the engineered features as well as the raw element. After identifying the feature leakage from “core_gsr_count”, the remaining features required further review. To distinguish genuine chemical separability from potential feature leakage, an ablation study was conducted across multiple baseline configurations, progressively removing engineered features

suspected of encoding the labels. Ultimately, it was determined that no additional features were causing leakage, and in fact were carrying meaningful signal as illustrated in the results.

A review of the preprocessing steps revealed an opportunity for possible improvement, specifically for the XGBoost model, which handles sparsity well. Instead of removing raw elements from the dataset based on the 1% threshold, all 89 raw elements were included in the final XGBoost model. As a result, there was a noticeable improvement on feature importance distribution, including higher reliance on raw elements such as molybdenum, mercury, and gadolinium, the last of which was not included in the baseline model.

Based on the model performance, there is no meaningful gap to be closed. The model is not overfitting or underfitting, it has essentially saturated the data. This means that traditional hyperparameter tuning like grid, random, Bayesian search over depth, learning rate, or regularization are likely to only produce changes in the fourth or fifth decimal place. Those changes won't be statistically distinguishable from noise across CV folds.

Neural Network (MLP)

The baseline MLP achieved near-perfect performance (99.90% accuracy, ROC-AUC 1.00) with a deliberately simple architecture: two hidden layers of 128 units, ReLU activation, dropout rate of 0.30, and Adam optimizer at learning rate 1e-3. Given this baseline performance, traditional hyperparameter tuning was conducted to confirm that

the needle would not be moved much by tuning, and to identify the final optimal configuration for downstream validation tasks.

A structured search was conducted over seven configurations varying four hyperparameters: learning rate (1e-3, 5e-4), dropout rate (0.2, 0.3, 0.4), hidden layer width (128, 256), and network depth (2, 3 layers). Each configuration was trained for up to 15 epochs with early stopping (patience = 5) on validation loss. The same group-aware train/validation/test split (60/20/20 via GroupShuffleSplit on stub id) and StandardScaler normalization from the baseline were preserved to ensure comparability. The *gsr_count* feature was excluded from all configurations due to the target leakage concern identified during baseline evaluation, leaving 56 input features.

Beyond traditional hyperparameter search, optimization efforts pivoted toward two forensically meaningful objectives:

- 1) Threshold Optimization. The default classification threshold of 0.5 is unlikely to be forensically optimal. In criminal proceedings, the costs of false positives (implicating an innocent person) and false negatives (missing critical evidence) are both severe but asymmetric depending on the use case. Precision-recall curves were used to identify three operating points: a high-specificity threshold that minimizes false positives for courtroom evidence, a balanced threshold that maximizes F1 score, and a high-sensitivity threshold that minimizes false negatives for investigative screening.
- 2) Uncertainty Quantification via Monte Carlo Dropout. Standard model predictions produce a single probability, but forensic analysts need to know how confident

the model is in that prediction. Monte Carlo Dropout (Gal & Ghahramani, 2016) enables uncertainty estimation by running multiple stochastic forward passes ($n=50$) with dropout active at inference time, producing a distribution of predictions for each particle. The standard deviation across passes serves as a measure of epistemic uncertainty. Particles with high variance, particularly those near the decision boundary, can be flagged for expert review rather than receiving a hard classification.

Results

Logistic regression

The tuned logistic regression model was evaluated on a held-out test set using the same group-aware split strategy. The model achieved:

- Accuracy: ~ 0.82
- ROC-AUC: ~ 0.90
- PR-AUC: ~ 0.88
- Precision: ~ 0.79
- Recall: ~ 0.85

These results indicate strong but not perfect separability between GSR and Non-GSR particles when dominant elemental features are removed. Compared to the baseline model, performance improvements from hyperparameter tuning were minimal, with accuracy, precision, recall, and F1-score remaining nearly unchanged.

The confusion matrix shows that the model maintains relatively high recall for GSR particles while still producing a non-trivial number of false positives and false negatives.

This reflects a more realistic classification scenario compared to the near-perfect baseline model, where performance was artificially inflated by dominant elemental signals.

The ROC curve demonstrates strong discriminative ability, with an AUC of approximately 0.90, while the precision-recall curve highlights solid performance on the minority class, particularly given the class imbalance.

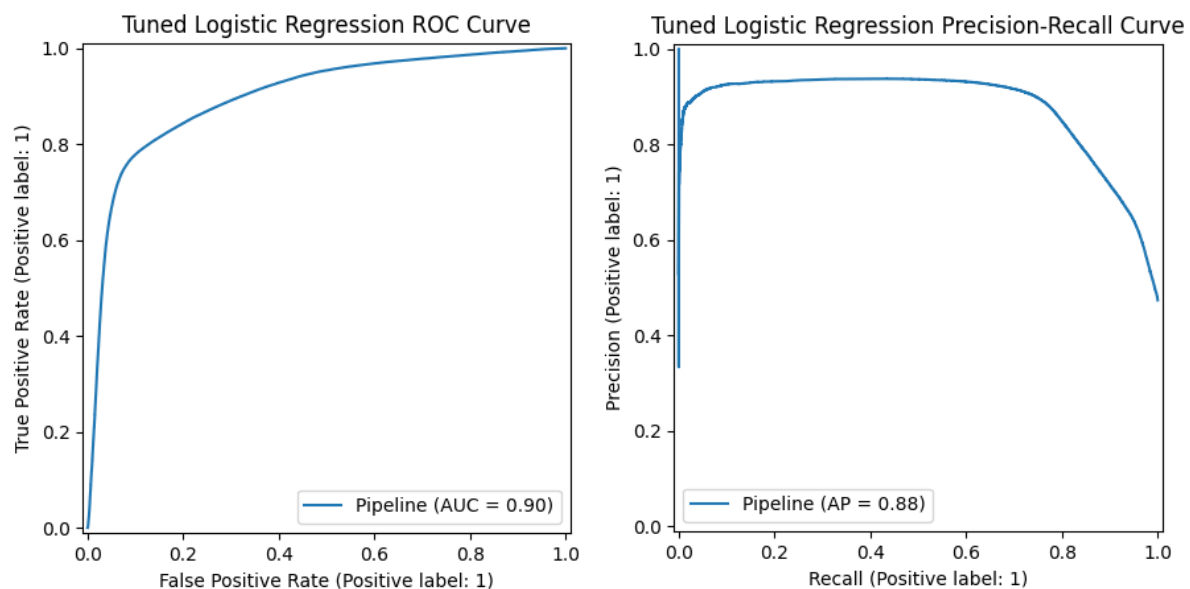


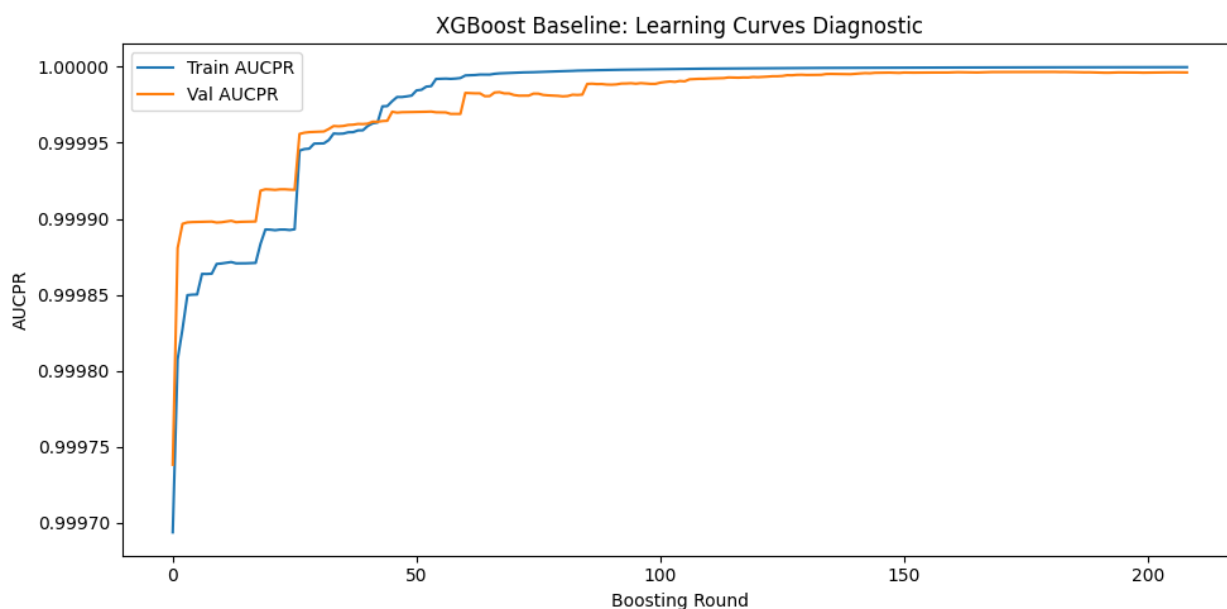
Figure 1 & 2: Tuned logistic regression ROC curve on the NFI test set ($AUC \approx 0.90$); Precision-recall curve for tuned logistic regression on the NFIR test set ($PR-AUC \approx 0.88$)

Overall, the tuned logistic regression model provides a meaningful benchmark for comparison with more complex models, capturing key patterns in the data while remaining interpretable.

XGB

The first step was to compare the model with and without the engineered features to identify whether any other features may potentially be causing leakage. All models at this point have removed the “core_gsr_count” feature, but v3 removes a few extra engineered features (as described in the baseline report) and v4 has no engineered features. Below are comparisons of the learning curves between the v2 model with engineered features (1st plot) and v4 with no engineered features (2nd plot).

Comparing the performance between the training and validation of plots 1 and 2, the evidence points toward true separability as opposed to leakage. The key factor here is that after removing the engineered features, the performance still remains high. The only features left in v4 are untransformed physical measurements, and the performance reflects the genuine separability of the underlying chemistry, not engineered artifacts. It is also worth noting that the v4 model (2nd plot) never reaches convergence. There is significant gap that never closes between training and validation, which further demonstrates that the convergence in the v2 model is likely due to legitimate signal from the engineered features.



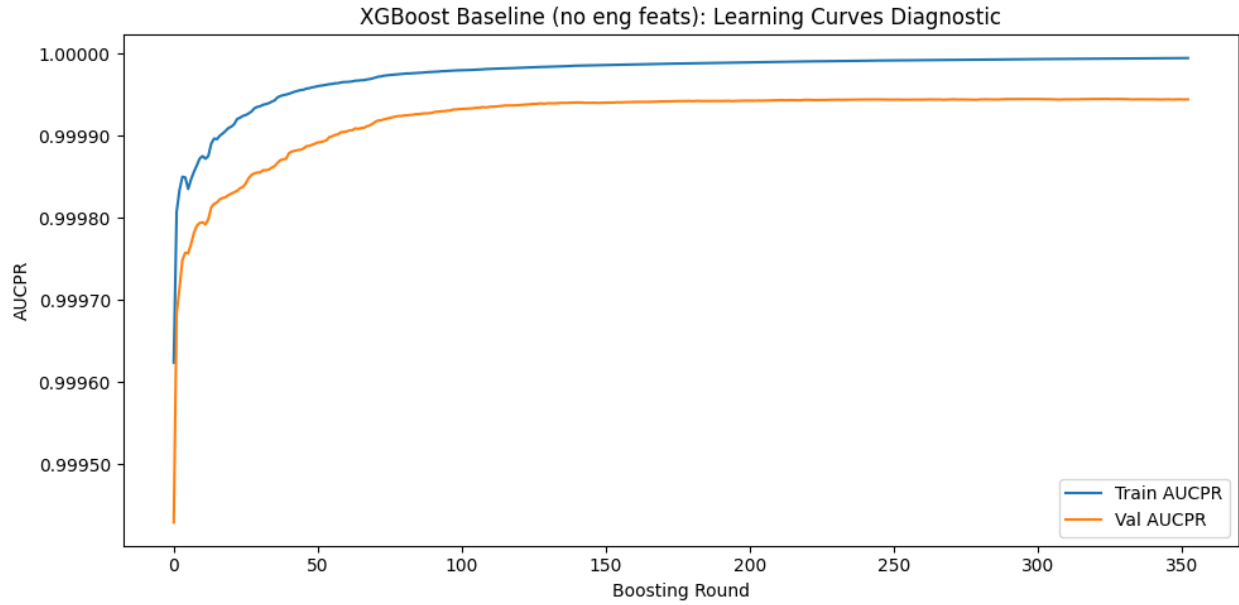


Figure 3 & 4: XGBoost learning curves: engineered vs. raw features

The “Baseline Model Comparison” figure below compares the summary metrics across v2, v3, and v4, which further strengthens this conclusion. If there were leakage rather than true separability, there would be a meaningful performance collapse at some point in the progression between v2 to v3 to v4, but instead the performance remains strong.

Baseline Model Comparison

Metric	Baseline v2	Baseline v3	Baseline v4
Features	40	37	27
Accuracy	0.999912	0.998998	0.998867
Precision (GSR)	0.999319	0.998886	0.999323
Recall (GSR)	0.999748	0.998999	0.998285
ROC-AUC	0.999997	0.99997	0.999953
PR-AUC	0.999996	0.99995	0.999952
False Positives	143	234	142
FPR	0.000613	0.001003	0.000609
Early Stopping Round	208	53	194
CV Mean PR-AUC	0.999998 ± 0.000001	0.999997 ± 0.000002	0.999997 ± 0.000002
CV Mean FPR	0.000525 ± 0.000216	0.000533 ± 0.000231	0.000533 ± 0.000231

Figure 5: XGBoost performance across feature sets (v2-v4)

After confirming that the remaining engineered features were carrying legitimate signal, the previously removed raw elements were reintroduced to the model. The 89 raw elements and 8 engineered features total to 97 feature columns in the final model. The learning curve diagnostic plot illustrates the same strong performance.

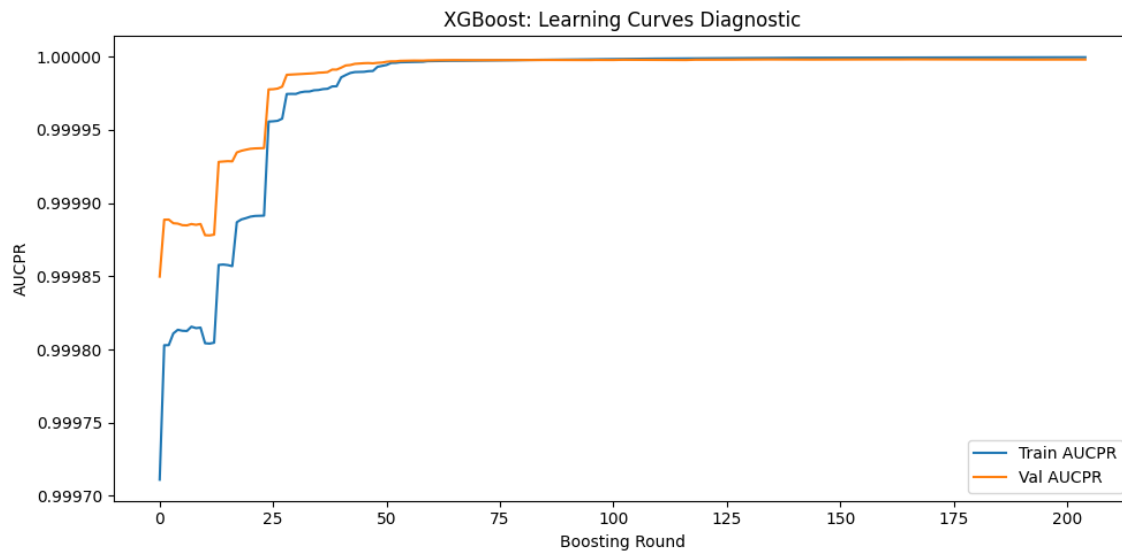


Figure 6: XGBoost learning curve showing train vs. validation AUC-PR over boosting rounds

The noticeable difference is the heavier reliance on some of the raw elements that were not previously included as important features in the baseline model such as molybdenum, mercury, and gadolinium.

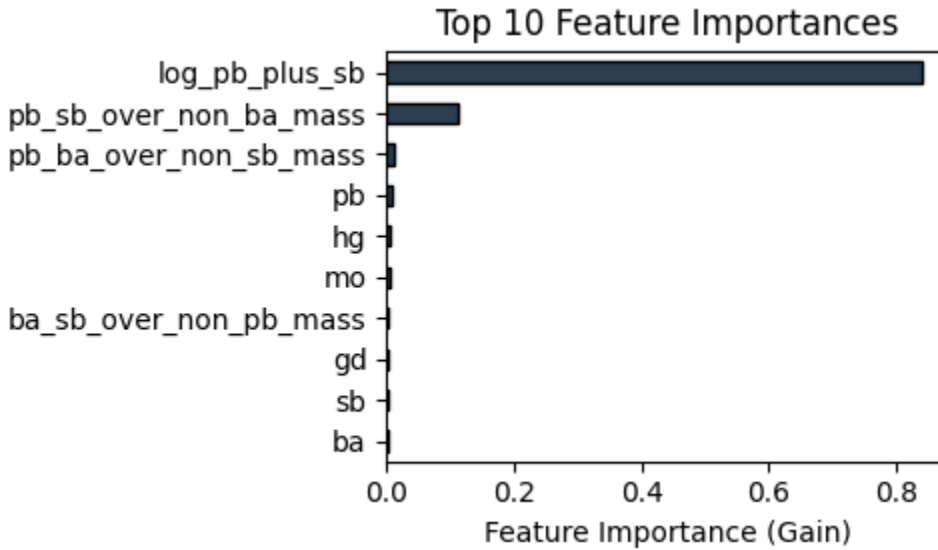


Figure 7: XGBoost features importance (final model)

Neural Network (MLP)

All seven configurations achieved ROC-AUC of 1.0000 and test accuracy above 99.93%, confirming that the binary GSR classification task is well-separated and that architecture choice is not the bottleneck. The top three configurations by validation loss were:

Table 1: Hyperparameter Search Results

Learning Rate	Dropout	Hidden Dim	Layers	Val Loss	Test FPR	Test Accuracy
0.001	0.3	256	2	0.001809	0.0010	99.93%
0.001	0.3	128	2	0.001815	0.0009	99.94%

0.0005	0.3	256	3	0.001838	0.0008	99.93
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The best configuration (hidden_dim=256, 2 layers, lr=0.001, dropout=0.3) was selected and retrained for up to 20 epochs with patience 5. The retrained model achieved a best validation loss of 0.001809. On the held-out test set, the tuned model produced:

- TN = 233,118 | FP = 157 | FN = 127 | TP = 209,822
- FPR: 0.07% | FNR: 0.06%
- ROC-AUC: 1.0000 | PR-AUC: 1.0000

Per-subclass analysis confirmed that the classic three-element GSR signature (PbBaSb) achieves perfect recall (1.0000), as does PbSb. The highest error rates remain concentrated in barium-containing classes: BaAl specificity of 0.9986 and BaSb recall of 0.9962, consistent with EDA findings that barium is the primary source of classification ambiguity.

The key takeaway from hyperparameter tuning is that traditional tuning does not meaningfully improve performance on this task. The marginal differences between configurations (val loss range: 0.0018–0.0022) are negligible in practical terms. This finding shifts the optimization focus from architecture search to threshold calibration and uncertainty quantification, which are more forensically relevant. Analysis of the precision-recall curve identified three operating points.

Table 2: Threshold Optimization Results

Mode	Threshold	FPR	FNR	F1	FP Count	FN Count
High-specificity	0.983	0.0086%	0.99%	0.9950	20	2,078
Balanced (best F1)	0.432	0.077%	0.045%	0.9993	180	95
Default (0.5)	0.500	0.067%	0.060%	0.9993	157	127
High-sensitivity	0.613	0.056%	0.099%	0.9992	130	208

At the high-specificity threshold (0.983), false positives drop from 157 to just 20, a 87% reduction, at the cost of 2,078 false negatives (0.99% FNR). For courtroom evidence where false positives carry the highest stakes, this operating point may be deemed appropriate. For investigative screening where missing GSR evidence is the primary concern, the default or balanced threshold provides better recall. Threshold analysis results can give forensic practitioners a tunable parameter rather than a fixed cutoff,

allowing them to calibrate the model's behavior to the specific use case.

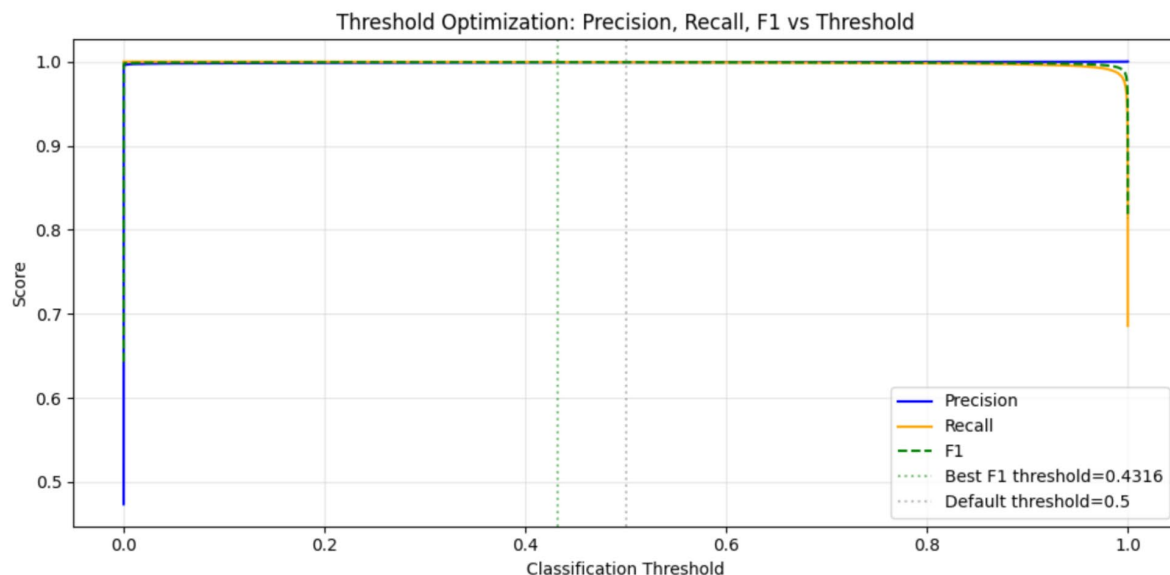


Fig. 8: Threshold Optimization Test

Across 50 stochastic forward passes on the test set (443,224 particles), the mean prediction standard deviation was 0.000762, indicating that the model is highly confident in the vast majority of its predictions. However, 1,761 particles (0.40%) exhibited uncertainty above 0.05, and 1,034 particles (0.23%) exceeded 0.10. Uncertainty by subclass revealed a pattern consistent with the false positive analysis.

Table 3: Monte Carlo Dropout Results.

Class	Label	N	Mean Std	Max Std	Std > 0.05
BaSb	GSR	20,708	0.0056	0.320	590
PbBa	GSR	48,129	0.0022	0.292	540
BaAl	Non-GSR	65,221	0.0012	0.329	496

BaCaSi	Non-GSR	81,503	0.0002	0.320	91
PbBaSb	GSR	108,047	0.0001	0.022	0
PbSb	GSR	33,065	0.0004	0.060	2

The model is highly confident on particles with the classic three-element GSR signature (PbBaSb: zero particles with std > 0.05). Uncertainty concentrates in barium-containing classes that sit near the decision boundary (BaSb, PbBa, and BaAl), which are particles that should be flagged for expert review. This validates the MC Dropout approach as a practical tool for identifying particles where automated classification should defer to human judgment.

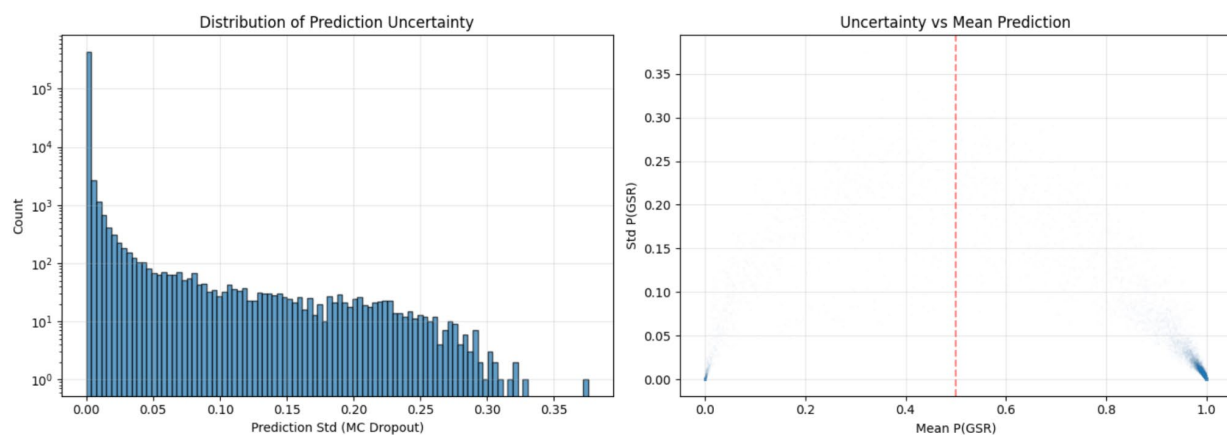


Fig. 9: Prediction Uncertainty Distribution (left, most are 0 but there is a long right tail) and Mean Prediction vs Uncertainty (right)

Feature Importance & Interpretability

Permutation importance was used to assess which features drive the tuned MLP's predictions. For each of the 56 input features, the feature's values were randomly shuffled across a 10,000-particle test sample (5 repetitions per feature), and the

resulting drop in ROC-AUC was measured. A large AUC drop indicates that the model relies heavily on that feature, a near-zero drop indicates the feature is redundant or uninformative for the classification task.

Features were also grouped by category, CLR-transformed elemental concentrations (27 features), binary presence indicators (27 features), and domain-engineered features (2 features), to assess which representation type contributes most to model performance. Finally, CLR and presence contributions were combined by element to identify which chemical elements are most important overall.

The permutation importance analysis revealed a striking concentration of predictive power in a very small number of features. The top 5 features by AUC drop are:

- Log_pb_plus_sb, mean AUC drop 0.0820
- Clr_sb, mean AUC drop 0.0006
- Has_ba, mean AUC drop 0.0002
- Clr_pb, mean AUC drop 0.0001
- Pb_sb_over_non_ba_o_mass, mean AUC drop 0.0001

Notably, the log_pb_plus_sb feature dominates all others by a factor of approximately 140x, accounting for the vast majority of the model's discriminative power. This feature captures the combined lead-antimony signal. The next most important features are the CLR-transformed antimony concentration, the binary barium presence indicator, and the CLR-transformed lead concentration, the three core GSR elements (Pb, Ba, Sb) that define the classic GSR signature.

By feature category, the two domain-engineered features (`log_pb_plus_sb` and `pb_sb_over_non_ba_o_mass`) account for 98.9% of total permutation importance (combined AUC drop of 0.082 out of 0.083 total). The 27 CLR-transformed concentration features contribute 0.97%, and the 27 binary presence indicators contribute 0.16%. This indicates that the neural network's hidden layers are primarily learning a refined, non-linear transformation of the hand-crafted domain features rather than discovering novel patterns in the raw compositional data.

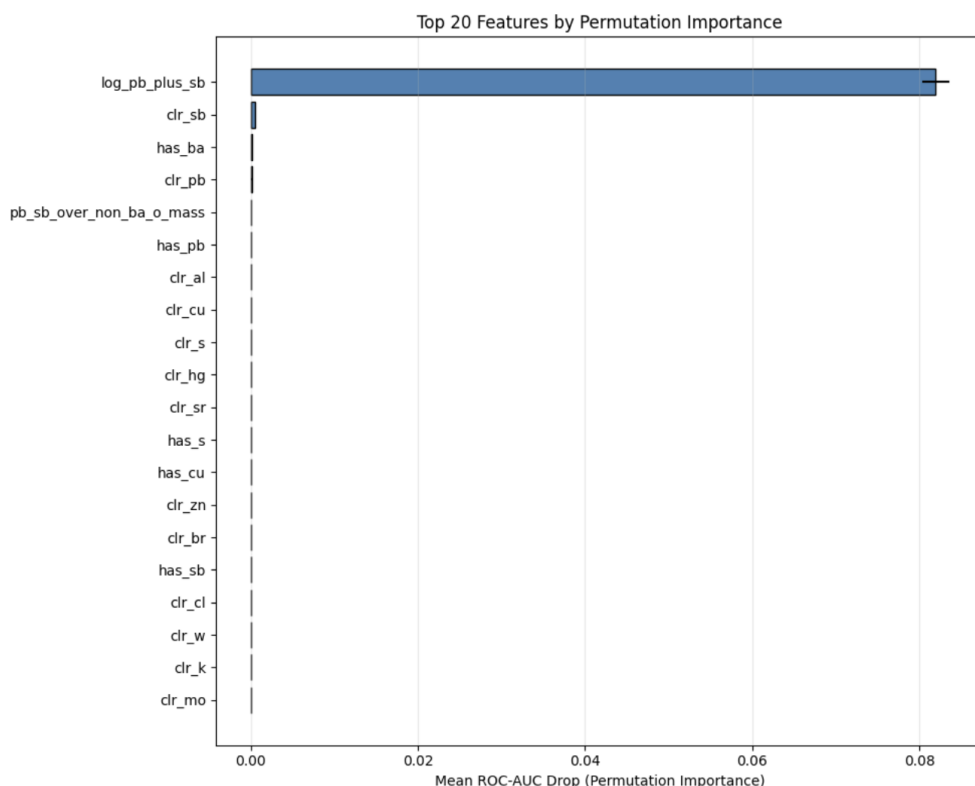


Fig. 10: Permutation based feature importance

When CLR and presence contributions are combined by element, the ranking aligns precisely with forensic domain knowledge:

- Sb (antimony) importance 0.000582

- Ba (barium) importance 0.000187
- Pb (lead) importance 0.000178
- Cu (copper) importance 0.000062
- Al (aluminum) importance 0.000049

The top three elements, antimony, barium, and lead, are the defining components of the classic GSR signature. Copper and aluminum appear in positions 4 and 5 because they are key indicators of non-GSR particle classes where copper and zinc co-occur in brass (CuZn class), and aluminum co-occurs with barium in environmental particles (BaAl class). The model uses these elements to distinguish true GSR from confounders that share individual elements with GSR but lack the full multi-element signature.

The feature importance analysis confirms that the MLP's learned decision boundary aligns with established forensic chemistry. The model is not relying on spurious correlations or artifacts in the training data, instead it is identifying particles based on the presence and relative concentration of lead, barium, and antimony, exactly as human forensic analysts do.

However, the extreme dominance of the `log_pb_plus_sb` engineered feature (98.7% of total importance) raises an important question about the model's added value. If a single hand-crafted ratio captures nearly all of the discriminative signal, the neural network's contribution is primarily in refining the decision boundary around edge cases, the barium-containing particles (BaAl, BaSb, PbBa) where the MC Dropout analysis identified elevated uncertainty. For the majority of particles with clear multi-element signatures, a simpler rule-based system using the Pb+Sb ratio would achieve comparable performance.

Targeted Ablation: Separability vs. Leakage Investigation

The baseline model achieved near-perfect classification performance (ROC-AUC $\geq$ 0.999). While this could reflect genuine separability in the data, it raises a legitimate concern of whether the models exploiting proxy information or engineered features that effectively encode the label, rather than learning generalizable chemical patterns.

Ablation Design

We trained the same MLP architecture under four conditions, progressively stripping away feature engineering to isolate the source of the signal.

- A: Full
CLR + presence + engineered
- B: CLR only
- C: Raw + presence
- D: Raw only

All conditions used the same group-aware train/val/test split training used a 10% subsample for efficiency, with the full test set (443,224 particles) for evaluation.

Table 4: Ablation Study Result by Condition

Condition	Features	ROC-AUC	Accuracy	FP	FN
Full	56	1.00	99.79%	871	74
CLR	27	0.99	99.74%	1,129	9

Raw + Presence	54	0.99	99.74%	1,109	31
Raw Only	27	0.99	99.61%	462	1,262

The critical finding is that even in condition D, the raw elemental concentrations only, with zero feature engineering, the model still achieves ROC-AUC of 0.9999 and 99.61% accuracy. This rules out feature engineering as the source of the near-perfect performance. The signal is in the raw data itself. This is expected given how the labels were constructed where we used the NFI labeling scheme to assign GSR vs Non-GSR based on the particle's elemental composition class, and the features are elemental compositions. The labels are a deterministic function of the feature space and a sufficiently flexible model will learn this mapping regardless of how the features are preprocessed. This is not leakage in the traditional sense where information from the test set contaminates training but rather true structural separability where the classification task is well-defined and the decision boundary is learnable from the raw measurements.

However, while overall discrimination is near-perfect across all conditions, the error profiles differ meaningfully. Condition D (raw only) produces 17× more false negatives compared to condition A (1,262 vs. 74, respectively). Without presence flags or engineered features, the model misses GSR particles that have low concentrations of signature elements.

Presence flags are the most impactful feature group. Adding them reduces false negatives from 1,262 to 31 (see condition $C \rightarrow D$). The binary "which elements are present" signal captures the co-occurrence patterns that define GSR more effectively than raw concentrations alone.

The CLR transform in comparison has minimal impact when presence flags are available (B vs. C are nearly identical). The NN can learn compositional relationships from either raw or CLR-transformed inputs. Engineered features provide only marginal improvement on the hardest cases, reducing FP from 1,129 to 871 when added to CLR+presence (B $\rightarrow$ A).

In conclusion, the ablation confirms that the binary GSR vs. Non-GSR classification task is genuinely well-separated in the elemental feature space, and that this separability is not an artifact of feature engineering. The more forensically relevant challenge is not improving overall accuracy (which is already at ceiling) but rather: (1) understanding model behavior on ambiguous single-element particles, (2) quantifying prediction uncertainty for edge cases, and (3) validating generalization to independent other datasets from different labs (e.g. NIST).

Cross-Dataset Validation on NIST Data

Logistic Regression

The logistic regression model demonstrated strong performance within the NFI data distribution, achieving an ROC-AUC of approximately 0.90 and PR-AUC of approximately 0.88 on the held-out NFI test set. These results indicate that a linear

model can effectively distinguish GSR from non-GSR particles when clear multi-element signatures are present.

To evaluate real-world forensic applicability, the model was tested on the independent NIST Gunshot Residue dataset, which represents a fundamentally different data distribution due to the differences in instrumentation, quantification methods, and laboratory protocols. No retraining or recalibration was performed; the NFI-trained model and preprocessing pipeline were applied directly to the NIST data.

Validation Strategy

The same NIST dataset was divided into:

- Non-shooter samples (95,447 particles): treated as a proxy for specificity (false positive analysis), including brake dust and firework residue
- Shooter samples (40,000 particles): used as a sensitivity contrast, expected to contain some true GSR particles

Model predictions were evaluated across multiple probability thresholds, including the default (0.50), a balanced/F1-optimized threshold (0.60), a high-specificity threshold (0.70), and a high-sensitivity threshold (~ 0.01), to assess operational trade-off.

Table 5: Threshold configurations derived from NFI performance for NIST evaluation

Mode	Threshold	FPR	FNR	Precision	Recall	F1	FP	FN
Default Threshold	0.50	0.203914	0.153613	0.788836	0.846387	0.816599	47,568	32,251
Balanced / Best F1	0.60	0.112087	0.210904	0.863688	0.789096	0.824709	26,147	44,279

High Specificity	0.70	0.057400	0.299392	0.916564	0.700608	0.794167	13,390	62,857
High Sensitivity	0.01	0.928213	0.003796	0.491335	0.996204	0.658094	216,529	797

Results

Non-Shooter Performance (False Positive)

On the NIST non-shooter dataset, the logistic regression model produced a measurable but controlled false positive rate:

- Default threshold (0.50): ~5.84% false positives (5,447 particles)
- Balanced threshold (~0.60): ~4.52% false positives
- High-specificity threshold (~0.70): ~4.66% false positives

False positives were not randomly distributed. Instead, they were concentrated in specific particle types:

- Firework-related particles contributed the majority (~10.95%)
- Brake dust particles contributed a smaller share (~1.22%)

Table 6: False positive rates across non-shooter particle sources

Source	Type	N	FP	FPR	Mean P(GSR)
brakes_chevy_front_driver	brakes	5000	11	0.0022	0.033487
brakes_chevy_front_passenger	brakes	5000	9	0.0018	0.032713

ssenger					
brakes_chey_rear_driver	brakes	5000	120	0.0240	0.072545
brakes_chey_rear_passenger	brakes	5000	58	0.0116	0.063811
brakes_ford_a213_front_driver	brakes	5000	63	0.0080	0.042910
brakes_ford_a213_front_passenger	brakes	5000	40	0.0278	0.030237
brakes_ford_a213_rear_driver	brakes	5000	139	0.0152	0.077027
brakes_ford_a213_rear_passenger	brakes	5000	76	0.0064	0.055517
brakes_ford_b297_front_driver	brakes	5000	32	0.0132	0.046957
brakes_ford_b297_front_passenger	brakes	5000	66	0.0000	0.055343

At the class level, the largest contributors included:

- Unclassified particles
- Uranium-bearing and iron-bearing particles

Table 7: False positive distribution across NIST particle classes.

Class	False Positives
Unclassified	5373
Uranium-bearing	96
Iron-bearing	68
Cu-rich	17
Ca-bearing	12
Iron	3
Silicate	2

This indicates that certain environmental particles exhibit partial chemical overlap with GSR signatures.

Shooter Performance (Sensitivity)

On the shooter dataset, the model identified relatively few particles as GSR:

- Default threshold (0.50): ~4.07% classified as GSR
- Balanced threshold (~0.60): ~3.47%
- High-specificity threshold (~0.70): ~2.92%
- High-sensitivity threshold (~0.01): ~20.34%

Table 8: GSR classification rates for shooter samples across sources.

Source	N	GSR	GSR Rate	Mean P(GSR)
shooter_1	5000	287	0.0574	0.068372
shooter_2	5000	346	0.0692	0.074454
shooter_3L	5000	281	0.0562	0.068484

shooter_3R	5000	224	0.0448	0.056941
shooter_4L	5000	45	0.0090	0.014894
shooter_4R	5000	47	0.0094	0.014171
shooter_5L	5000	146	0.0292	0.033115
shooter_5R	5000	251	0.0502	0.059608

Even under more permissive thresholds, the model assigns low GSR probabilities to the majority of shooter particles.

Probability Distribution Comparison

The predicted probability distributions for shooter and non-shooter particles show substantial overlap:

- Mean $P(\text{GSR})$, non-shooter ≈ 0.088
- Mean $P(\text{GSR})$, shooter ≈ 0.085
- GSR classification rate at default threshold:
 - Non-shooter: 5.84%
 - Shooter: 4.07%

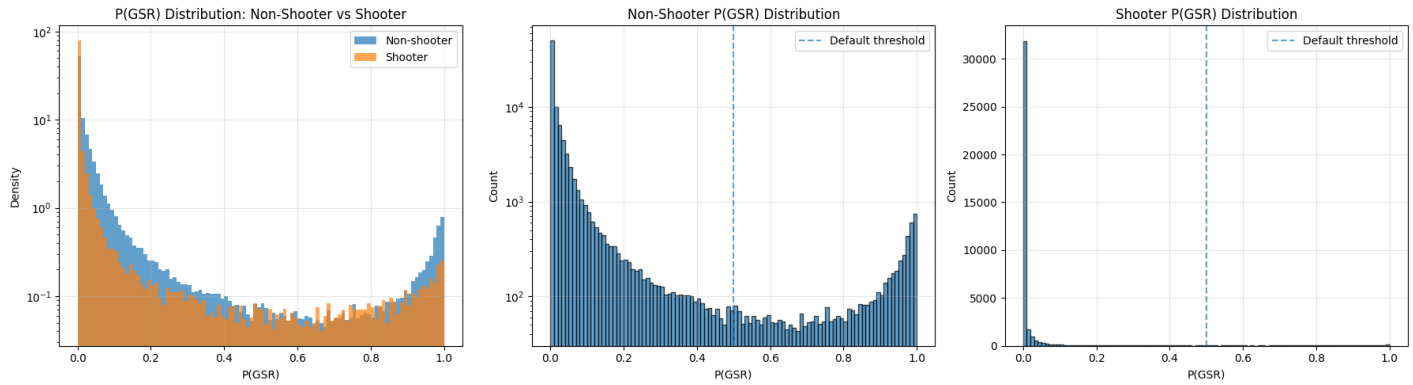


Fig. 11: Predicted GSR probability distributions for NIST shooter and non-shooter particles

The expected contrast—higher GSR probabilities for shooter particles—does not materialize. Instead, both groups are concentrated at low predicted probabilities with minimal separation.

Interpretation

The NIST validation demonstrates that the logistic regression model does not generalize effectively to out-of-distribution data from a different laboratory. While the model captures meaningful signal within the NFI dataset, its learned decision boundary does not transfer to NIST data.

Rather than over-classifying particles as GSR, the model exhibits a conservative bias, assigning low probabilities across both shooter and non-shooter samples. This results in:

- Low false positive rates relative to more flexible models
- But also low sensitivity, limiting its ability to detect true GSR in unseen data

The lack of separation between shooter and non-shooter probability distributions suggests that the model is strongly dependent on NFI-specific feature distributions. Differences in measurement methodology introduce a distribution shift that cannot be resolved through threshold adjustment alone.

Key Takeaways

- Logistic regression achieves strong in-distribution performance but fails to generalize across laboratories
- The model is conservative, producing relatively low false positives but also low sensitivity
- False positives are systematic and source-dependent, particularly in firework-related particles
- Threshold tuning adjusts trade-offs but does not resolve the underlying overlap between classes
- These findings highlight that feature representation and cross-laboratory variability, rather than model complexity alone, are primary limitations to generalization

XGB

The NIST dataset can be used for crosstesting to evaluate the generalizability of the XGBoost model. The preprocessing steps for the XGBoost model in regards to the NIST dataset were slightly different than the other models because the final model reintroduced all 89 raw elements. While other models may have reduced the NIST data down to the 27 informative elements, the XGBoost model did not have to reduce the

NIST raw elements. Instead, it reduced the 89 NFI raw elements down to match the total number of NIST raw elements, since all NIST raw elements exist within the NFI's 89 initial element list.

From there, several more things needed to be done for preprocessing the NIST dataset.

The appendix in the EDA report noted several significant differences between the NFI and NIST datasets. Firstly, the elemental weights in comparison to oxygen needed to be renormalized. The XGBoost steps for this can be seen in the Appendix E notebook.

The NIST dataset has a variety of sample sources, some are Non-GSR confounders like fireworks or brake pads, while others are GSR shooter sources. The particles from Non-GSR sources were all assigned 0 as the target since they are guaranteed not to be from a shooter source. These Non-GSR sources were then crosstested directly for model performance on false positive rate because in theory any particle identified as GSR by the model from a Non-GSR source will be a false positive. The results of this testing are below.

NIST Non-GSR validation

Total particles: 105,246

TN (correctly Non-GSR): 48,365

FP (flagged as GSR): 56,881

FPR: 0.540458

As these numbers show, the XGBoost model was a little worse than a 50/50 guess in terms of false positives. The number should be 0, but instead over half of the particles from the NIST non-GSR sources were identified as GSR. Clearly, there are numerous

problems with the model generalizing to datasets outside of the NFI data, which will be discussed in the “Discussion and Next Steps” section for XGBoost.

Neural Network (MLP)

The NFI-trained models demonstrated strong performance within the NFI data distribution, but forensic applicability demands generalization beyond a single laboratory's data. The NIST Gunshot Residue Dataset (Ritchie & Reynolds, 2021) provides an independent source of particle measurements from known sample types, including shooter hand swabs, brake dust from multiple vehicles, and firework residue, making it an ideal out-of-distribution test bed.

This validation step directly addresses the most forensically relevant question: do a model's learned decision boundary transfer to particles measured by a different laboratory, with different instruments, different quantification methods, and different sample preparation protocols? The following sections describe our methods and results.

NIST Data Preparation

The NIST data was accessed from the concatenated neQuant parquet files. The neQuant quantification method was selected because it is the only method present across 100% of the 30 NIST sample sources. The concatenated dataset contains 135,447 particles across 30 sources (22 non-shooter, 8 shooter).

Several preprocessing steps were required to align the NIST data with the NFI-trained model's expected input format. The NIST dataset contains 36 raw elemental columns compared to the NFI's 27 informative elements. Five NFI informative elements are absent from NIST (Na, As, Br, W, Hg), and fourteen NIST-specific elements not in the

NFI informative set were dropped. Uncertainty columns, imaging metadata, and auxiliary columns were also removed.

A critical difference between the datasets is how oxygen is represented. In the NFI data, elemental weight percentages include oxygen and sum to approximately 100%. In the NIST data, elemental weights sum to 100% excluding oxygen, with oxygen reported as an additional value. To align conventions, NIST elemental values were renormalized by dividing each element's weight by the total sum with oxygen and multiplying by 100.

An initial validation attempt using all 27 NFI elements (with the 5 missing elements zero-filled) produced a 77% false positive rate on known non-GSR confounders. Diagnostic analysis revealed that the zero-filled elements severely distorted the CLR transform and binary presence features, creating a systematic distribution shift. To address this, a fair comparison model was retrained using only the 22 elements shared between both datasets, with the same architecture and training procedure. This shared-element model achieved comparable NFI test performance (ROC-AUC 1.0000, FPR 0.0009) while ensuring both datasets were on equal footing.

Validation Strategy

We assigned binary GSR vs. non-GSR ground truth based on non-shooter samples (95,447 particles from 22 sources) that include brake dust from multiple vehicles and firework residue from various stages of handling and ignition versus shooter samples (40,000 particles from 8 sources) from the hands of individuals who discharged firearms and should contain a mixture of true GSR and environmental background particles. We

expect the model to classify a substantially higher proportion of shooter particles as GSR compared to non-shooter particles.

Results

The shared-element MLP (22 elements, 46 features) was applied to both NIST subsets using the NFI-fitted scaler and preprocessing pipeline. No NIST data was used to fit any transformations.

The model classified 73.7% of known non-GSR confounder particles as GSR at the default 0.5 threshold. Even at the high-specificity threshold (0.983), the false positive rate remained at 61.5%. By source type, brake dust particles triggered a higher false positive rate (81.5%) than firework residue (65.1%). The most common false positive classes were Unclassified (36,227), Iron-bearing (13,001), and Ca-bearing (6,995).

The model classified 63.4% of shooter particles as GSR at the default threshold. This is actually lower than the non-shooter false positive rate (73.7%), meaning the model does not meaningfully distinguish between shooter and non-shooter NIST particles. The expected contrast did not materialize. The figure below illustrated this. The left panel shows overlaid density histograms of non-shooter (blue) and shooter (red) $P(\text{GSR})$ distributions on the log scale. The key observation is that both distributions look nearly identical, with the majority of mass above 0.5. There is no clear separation between the two. The middle panel shows the non-shooter $P(\text{GSR})$ histogram where most particles cluster at high probabilities (above the 0.5 threshold line). Finally, the right panel shows the shooter $P(\text{GSR})$ histogram. We see that the NFI trained model fails to generalize

and can not distinguish NIST non-shooter from NIST shooter since it is treating all NIST particles as out-of-distribution relative to NFI.

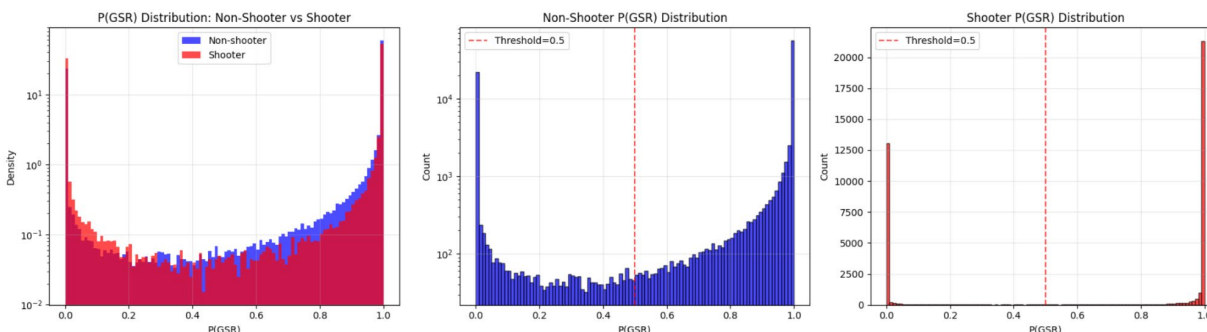


Fig. 12: $P(\text{GSR})$ distributions from NFI trained model on NIST data

The NIST validation reveals that the NFI-trained MLP does not generalize to out-of-distribution data from a different laboratory. The model classifies the majority of all NIST particles as GSR regardless of source type, indicating that it learned NFI-specific compositional distributions rather than generalizable GSR chemistry.

Diagnostic analysis comparing scaled feature distributions between NFI and NIST data identified pervasive distribution shift: 9 of 46 shared features had mean differences exceeding 2 standard deviations, with the maximum shift of 7.68 standard deviations occurring in the strontium presence indicator. This shift is driven by fundamental differences in measurement methodology between the two laboratories from using different instruments, calibration protocols, quantification methods, and reference spectra from laboratories in two different countries, rather than by the model architecture or training procedure.

This finding has direct implications for forensic deployment. A model trained on one laboratory's data cannot be blindly applied to another laboratory's data without domain adaptation. The 99.9% accuracy on NFI data reflects the model's ability to replicate NFI's classification protocol with high fidelity, but that protocol is instrument-specific. Cross-laboratory generalization requires either (1) retraining on the target laboratory's data, (2) domain adaptation or transfer learning techniques, (3) training on combined multi-laboratory datasets, or (4) standardized measurement protocols across laboratories.

Importantly, this limitation is not unique to the MLP or to our specific implementation. It reflects a fundamental challenge in forensic GSR analysis where the lack of standardized measurement protocols across laboratories means that any supervised model trained on single-source data will face the same generalization barrier. This finding underscores the need for multi-laboratory validation as a prerequisite for any automated GSR classification system intended for real-world deployment.

4. Discussion & Next Steps

Logistic regression

The original analysis plan for the logistic regression track assumed that performance improvements would primarily come from hyperparameter tuning and regularization adjustments, including exploration of penalty types (L1, L2, elastic net) and the strength of regularization ("C"). However, the baseline and tuned model results demonstrated that these adjustments yielded only marginal performance gains, with ROC-AUC stabilizing around ~0.90 and PR-AUC around ~0.88. This indicates that, similar to more

complex models, logistic regression performance is not constrained by model specification, but rather by the underlying feature representation.

This finding redirected focus away from model tuning and toward feature engineering and threshold calibration. Threshold analysis showed that modest improvements in precision and false positive control can be achieved by increasing the classification threshold (e.g., from 0.50 to ~0.60-0.70), though this comes at the cost of reduced recall. These trade-offs are operationally meaningful in a forensic setting, where minimizing false positives may be prioritized over maximizing sensitivity depending on the use case. However, threshold tuning alone does not materially change the model's overall discriminative ability, reinforcing that feature quality, not threshold selection, is the primary driver of performance.

The most significant deviation from the original analysis plan emerged during the NIST cross-dataset validation. The expectation was that the NFI-trained logistic regression model would generalize to the NIST dataset with some degradation, but still retain meaningful separation between shooter and non-shooter particles. Instead, the model exhibited limited generalization, producing low GSR classification rates for both groups and failing to create meaningful separation in predicted probabilities. At the default threshold, non-shooter particles were classified as FSR at a rate of ~5.84%, while shooter particles were classified at an even lower rate of ~4.07%, indicating that the model does not effectively distinguish between the two groups in the NIST domain.

This behavior contrasts with overfitting-driven failure modes and instead reflects a distribution shift problem. The model-learned decision boundaries specific to the NFI

measurement space and applies them conservatively when faced with out-of-distribution data. Rather than over-classifying GSR, the logistic regression model defaults toward low predicted probabilities, suggesting that it cannot confidently map NIST particle compositions into the learned GSR/non-GSR boundary.

The false positive analysis further supports this interpretation. Misclassifications are not random, but concentrated in specific environmental particle types, particularly firework-related particles and, to a lesser extent, brake dust. These particles likely share partial compositional overlap with GSR, highlighting a key limitation of the current feature set: it captures strong signal within the NFI dataset but does not fully resolve chemical ambiguity across domains.

Feature importance results from the logistic regression model reinforce a critical insight for future modeling. A small subset of elemental features (notably those related to Pb, Sb, and Ba in earlier iterations) drives a disproportionate share of model performance. While these features provide strong discriminatory power within NFI data, they are also highly sensitive to instrument calibration and measurement differences, which likely contributes to the observed breakdown in cross-laboratory generalization. The removal of these dominant elements in the final model reduces artificial separability but exposes the underlying challenge of distinguishing GSR from chemically similar particles when dominant signals are absent or less reliable.

These findings suggest that future improvements should prioritize feature representation over model complexity. In particular, ratio-based features (e.g., Pb/Ba, Pb/Sb, Ba/Sb, or normalized elemental proportions) may provide greater robustness across laboratories

than absolute or CLR-transformed concentrations. Ratios between elements measured within the same sample may be less sensitive to inter-laboratory calibration differences and therefore better suited for cross-dataset generalization.

Additionally, domain adaptation strategies represent a promising next step. Fine-tuning the logistic regression model on a small labeled subset of NIST data could recalibrate the decision boundary to the new measurement domain. Given the relative simplicity and interpretability of logistic regression, this approach may offer a practical and efficient pathway for adapting models to new laboratory environments without requiring full retraining.

For cross-laboratory deployment, the NIST validation framework established in this analysis should be treated as a required evaluation standard. Key metrics include:

- False positive rate on non-shooter samples (target: minimal, ideally <1%)
- GSR classification rate contrast between shooter and non-shooter samples
- Degree of separation between predicted probability distributions.

The current results indicate that, while logistic regression is effective for within-laboratory classification, it is insufficient for direct cross-laboratory application without additional adaptation.

Overall, the logistic regression analysis confirms that:

- Model performance is primarily driven by feature engineering rather than model tuning
- Cross-dataset generalization is limited by distribution shift, not model complexity

- False positives are systematic and chemically interpretable, not random
- Robust deployment requires domain-aware feature design and validation on target data

These insights define the next phase of the project, which should focus on developing cross-laboratory stable features, evaluating ratio-based representations, and exploring lightweight domain adaptation approaches to improve generalization beyond a single measurement environment.

XGB

The model is performing well enough on the NFI dataset that hyperparameter tuning and optimization are not a priority. The most glaring issue is its inability to generalize to a dataset outside of the NFI data. In the previous baseline report, a list of assumptions were discussed that included concerns about data collection differences and hardware configuration differences that could not be accounted for. Although we still cannot confirm this, this is likely a significant contributor as to why the model did not generalize well.

However, another possibility should be discussed. Looking at the NFI data, it appears that all source samples are from shooters, or at least that is what is inferred from the repository descriptions. If that's the case, then the "GSR" vs. "Non-GSR" labels actually only apply to shooter sources. Therefore, the XGBoost model was trained on shooter-only data and not actual confounder sources like the NIST dataset has.

With this in mind, two specific steps are considered for follow-up evaluation. First, the XGBoost model could be crosstested against the NIST shooter source data. This wouldn't be reliable enough to make the model usable in the court of law, but it would

help determine if this theory has any validity. To do this, major assumptions would need to be made with regards to labels. The NIST class names are not 1-to-1 with the NFI dataset, so label assignment is more complicated. For example, with NIST data, classes that begin with “GSR” are intended to describe particles that are characteristic of GSR, but not necessarily a positively identified GSR particle. This is known because there are particles in the Non-GSR sources that have GSR class names, but these are the confounder particles that cause false positives, not actual GSR particles. So, NIST shooter sources have GSR-class particles which would be assumed to be labeled “GSR” since they are in fact from a GSR source. This is a major assumption to make but will be necessary for this cross testing. After label assignment on the shooter source data, the trained model could be tested against NIST shooter-only data to evaluate how it does, since it appears to only have been trained on shooter data.

Secondly, the XGBoost model could be rebuilt from scratch on the NIST data. All Non-GSR sources are known to be Non-GSR, including the GSR classes which are in fact confounders. The shooter source data has the labels that were assumed as described above. After training the model on both shooter and non-shooter sources, the model would, in theory, be more capable of generalizing to secondary data sources since it has been trained on both shooter and non-shooter data.

If the above theory holds any truth, this would explain why the NFI data was trivial to the non-linear models. There was no real confounder source in the NFI model and therefore the models were only learning that particles high in lead and antimony were GSR particles from a GSR source, which is a very obvious statement to make. The challenge would have been to learn true confounder source data in addition to the shooter source

data, and try to differentiate between a GSR-characteristic particle from a Non-GSR source (true confounder) versus a GSR-characteristic particle from a GSR source (true GSR). These next steps will be conducted and the results compiled in the final report.

Neural Network (MLP)

The original analysis plan anticipated that hyperparameter tuning would be the primary optimization lever for the neural network track, with the expectation that architecture search over learning rates, dropout rates, hidden dimensions, and network depth would yield meaningful performance improvements. The baseline results report quickly invalidated this assumption. In the final models tuning experiments, all seven configurations in the hyperparameter search achieved ROC-AUC of 1.0000 and test accuracy above 99.93%, with validation loss differences on the order of 0.0004 which is negligible in practical terms. The binary GSR classification task on NFI data is sufficiently well-separated that model architecture is not the bottleneck.

This finding redirected the optimization effort toward forensically meaningful objectives, threshold calibration and uncertainty quantification. The threshold analysis revealed that false positives can be reduced by 87% (from 157 to 20) by raising the classification threshold to 0.983, at the cost of a 0.99% false negative rate, a tradeoff that forensic practitioners can calibrate to their specific use case. The Monte Carlo Dropout analysis identified 1,761 particles (0.40% of the test set) with meaningful prediction uncertainty, mainly concentrated in the barium-containing classes (BaSb, PbBa, BaAl) that the baseline error analysis had flagged. These are particles that should be deferred to expert review rather than receiving automated classification.

The most significant change to the analysis plan came from the NIST cross-dataset validation. The original plan treated NIST validation as a confirmatory step with the expectation that the NFI-trained model would generalize to NIST data with some degradation but still demonstrate meaningful discrimination. Instead, the model failed catastrophically, classifying 73.7% of known non-GSR confounder particles as GSR and showing no meaningful contrast between shooter and non-shooter samples. This result indicates that while the MLP successfully automates NFI's classification protocol with 99.9% fidelity, the protocol is instrument- and laboratory-specific. The model learned NFI's compositional patterns, not generalizable GSR chemistry.

TabNet was originally planned as an additional model for its built-in attention masks, which provide per-particle feature importance. However, the permutation importance analysis revealed that a single engineered feature ($\log_{pb_plus_sb}$) accounts for 98.9% of the model's discriminative power. This concentration of importance in one hand-crafted ratio diminishes the value of TabNet's attention mechanism as there is little ambiguity about what the model is attending to. The feature importance results also suggest a more impactful modeling direction. Given that the Pb+Sb ratio dominates predictions, a simpler model built around instrument-invariant elemental ratios may generalize better across laboratories than the full CLR-based pipeline. Specifically, ratio-based models using features like Pb/Ba, Pb/Sb, Ba/Sb, and (Pb+Sb) divided by total_mass, rather than CLR-transformed concentrations, may be more robust to the calibration differences that caused the NIST generalization failure. Ratios between elements measured by the same instrument should be more stable across laboratories

than absolute concentrations, which depend on instrument-specific calibration and quantification methods.

Fine-tuning the NFI-trained MLP on a small labeled subset of NIST data (transfer learning) remains a viable path. The feature importance analysis suggests that recalibrating the Pb+Sb ratio threshold for the NIST measurement domain could be sufficient, since the model's decision boundary is primarily defined by this single feature.

Any model intended for cross-laboratory deployment must be validated on the target laboratory's data. The NIST validation framework established in this analysis provides a reusable evaluation protocol. The key metrics are:

1. False positive rate on non-shooter NIST samples (ideally below 1%)
2. GSR classification rate contrast between shooter and non-shooter samples (at least 5x ratio)
3. Probability distribution separation between the two groups (target: bimodal distribution with clear separation at the decision boundary)

For within-NFI evaluation, the existing framework (group-aware cross-validation, per-subclass analysis, threshold optimization, MC Dropout uncertainty) is sufficient and should be applied consistently across all model architectures.

For NFI-only models, minimal additional tuning is needed. The hyperparameter search demonstrated that the task is not sensitive to architecture choices, and the feature importance analysis confirmed that performance is driven by the engineered features rather than the network's capacity to learn complex interactions from raw data. But for cross-laboratory models, the key parameters to investigate are:

1. Which elemental ratios are most stable across NFI and NIST measurement domains?
2. The Pb+Sb ratio threshold that separates GSR from Non-GSR in NIST data (compared to the NFI-learned threshold)
3. The minimum amount of labeled NIST data needed for effective fine-tuning

These represent the next phase of the project and would benefit from a collaborative arrangement with NIST to establish ground-truth GSR labels for their shooter samples.

Citations

Matzen, T., Kukurin, C., van de Wetering, J., Ariëns, S., Bosma, W., Knijnenberg, A., ... & Ypma, R. J. (2022). Objectifying evidence evaluation for gunshot residue comparisons using machine learning on criminal case data. *Forensic science international*, 335, 111293.

Ritchie, Nicholas W. M., Renolds, Amy (2021), Automated particle analysis (SEM/EDS) data from samples known to have been exposed to gunshot residue and from samples occasionally mistaken for gunshot residue - like brake dust and fireworks., National Institute of Standards and Technology, <https://doi.org/10.18434/mds2-2476> (Version: 1.0, Accessed 2026-03-21)

Appendices

Appendix A: GitHub Repository (*Team Delta Capstone*)

<https://github.com/bkoconnell/datascience-capstone>

Appendix B: Data Dictionary XGBoost

https://github.com/bkoconnell/datascience-capstone/blob/artifacts/artifacts/reports/04_model_tuning_and_optimization/data_dictionaries/xgb_model_datadict.md

Appendix C: Data Dictionary Neural Network

The neural network feature dataset (engineered_features_nn.parquet) contains 2,294,985 rows and 62 columns.

Metadata Columns

Column	Type	Description
stub_id	int	SEM stub identifier
particle_id	int	Particle identifier within stub
label	string	"GSR" or "Non_GSR"
target	int	Binary target: GSR = 1, Non_GSR = 0
final_class	string	Particle subclass (e.g.,

		PbBaSb, CuZn, BaAl)
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CLR-Transformed Element Features (27 columns)

Each column is the Centered Log-Ratio transform of the corresponding element's weight-percent concentration. Values are real-valued and unconstrained (can be negative). Naming convention: *clr_{element}* .

Column	Element
clr_o	Oxygen
clr_cl	Chlorine
clr_cr	Chromium
clr_s	Sulfur
clr_k	Potassium
clr_br	Bromine
clr_cu	Copper
clr_na	Sodium
clr_mo	Molybdenum
clr_ba	Barium
clr_mg	Magnesium
clr_sr	Strontium
clr_al	Aluminum
clr_ti	Titanium
clr_ni	Nickel
clr_si	Silicon

clr_sn	Tin
clr_w	Tungsten
clr_ca	Calcium
clr_p	Phosphorus
clr_hg	Mercury
clr_pb	Lead
clr_mn	Manganese
clr_sb	Antimony
clr_as	Arsenic
clr_fe	Iron

Binary Presence Features (27 columns)

Each column is 1 if the corresponding element has concentration > 0, and 0 otherwise.

Naming convention: *has_{element}* .

Engineered Features (3 columns)

Column	Formula	Description
pb_sb_over_non_ba_o_mass	$(\text{Pb} + \text{Sb}) / (\text{total_mass} - \text{Ba} - \text{O})$	Mass fraction of Pb+Sb excluding Ba and O; sentinel value -1 when denominator = 0
log_pb_plus_sb	$\log_{10}(\text{Pb} + \text{Sb})$	Log-transformed additive GSR signal
gsr_count	count of {Pb, Ba, Sb} > 0	Number of core GSR

		elements present (0–3)
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Appendix D: Data Dictionary Logistic Regression

Metadata Columns

Column	Type	Description
stub_id	int	SEM stub identifier
particle_id	int	Particle identifier within stub
label	string	“GSR”, “Non_GSR”, or “Ambiguous”
target	int	Binary target: GSR = 1, Non_GSR = 0

Log-Transformed Element Features

Column	Description
log_pb	Log-transformed lead concentration
log_ba	Log-transformed barium concentration
log_sb	Log-transformed antimony concentration

Ratio Features

Column	Formula	Description
log_pb_sb_ratio	$\log(\text{Pb} / \text{Sb})$	Relative relationship between Pb and Sb
log_pb_ba_ratio	$\log(\text{Pb} / \text{Ba})$	Relative relationship

		between Pb and Ba
log_ba_sb_ratio	$\log(\text{Ba} / \text{Sb})$	Relative relationship between Ba and Sb

GSR vs Confounder Features

Column	Formula	Description
log_pb_zn_ratio	$\log(\text{Pb} / \text{Zn})$	Distinguishes GSR from environmental particles
log_sb_cu_ratio	$\log(\text{Sb} / \text{Cu})$	Separates GSR from copper-rich particles
log_ba_ti_ratio	$\log(\text{Ba} / \text{Ti})$	Differentiates industrial/environmental sources
log_gsr_over_confounders	Combined ratio	Overall GSR signal vs confounders

Composition Features

Column	Description
pb_pct	% of particle mass from Pb
ba_pct	% of particle mass from Ba
sb_pct	% of particle mass from Sb

Interaction Features

Column	Description
pb_sb_interaction	Interaction between Pb and Sb
pb_ba_interaction	Interaction between Pb and Ba

ba_sb_interaction	Interaction between Ba and Sb
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Appendix E: XGBoost Steps for NIST Cross Testing

Renormalize the Oxygen:

https://github.com/bkoconnell/datascience-capstone/blob/main/notebooks/99_sandbox/validation/renormalize_NIST_oxygen.ipynb

Cross-Validation Test with NIST:

https://github.com/bkoconnell/datascience-capstone/blob/main/notebooks/06_validation/xgb_nist_validation.ipynb